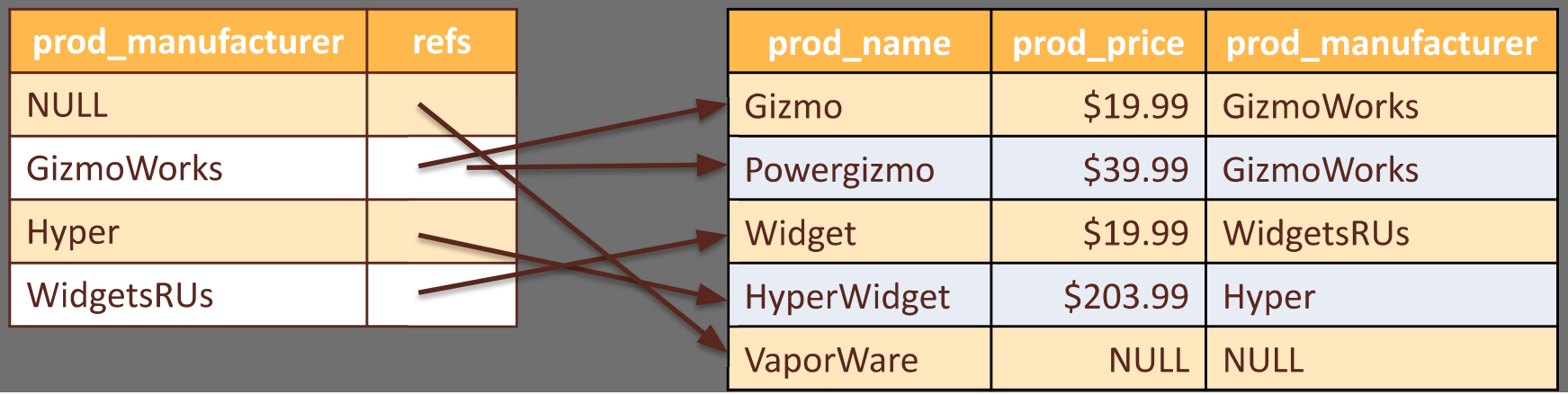
Index Concepts

What is an index?

A mapping from search key to tuple references.

Values of an attribute set

E.g., disk or memory location.
Possibly just to start of page.



prod_manufacturer	refs
NULL	
GizmoWorks	
Hyper	
WidgetsRUs	

prod_name	prod_price	prod_manufacturer
Gizmo	$19.99	GizmoWorks
Powergizmo	$39.99	GizmoWorks
Widget	$19.99	WidgetsRUs
HyperWidget	$203.99	Hyper
VaporWare	NULL	NULL

Index operations

- Search
- Insert
- Delete

} Preferably in bulk

What is the search key?

Often, but not required:

- Primary key
- A single attribute

Is it a key?

- Yes: Index maps each search key value to 1 location.
- No: Index maps each search key value to 1+ location.

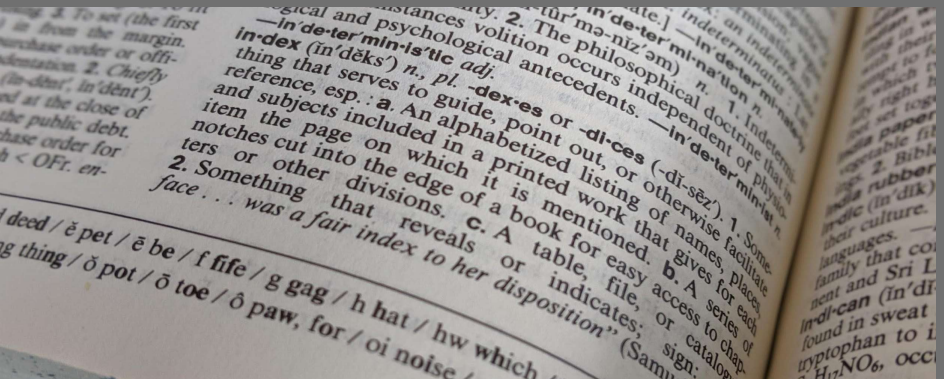
Primary vs secondary index

Primary / Clustered

- Determines storage order
- Typically PK
- Not separate data structure
- At most one

Secondary

- Doesn’t determined order
- Any search key allowed
- Separate data structure
- Multiple allowed



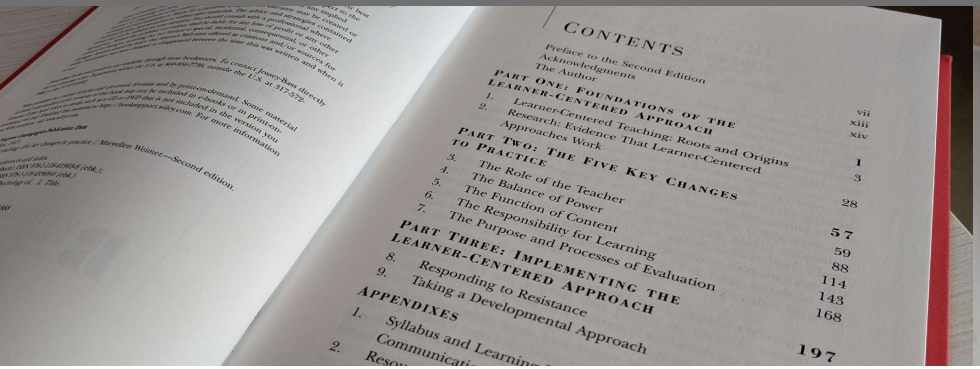
Sparse vs dense index

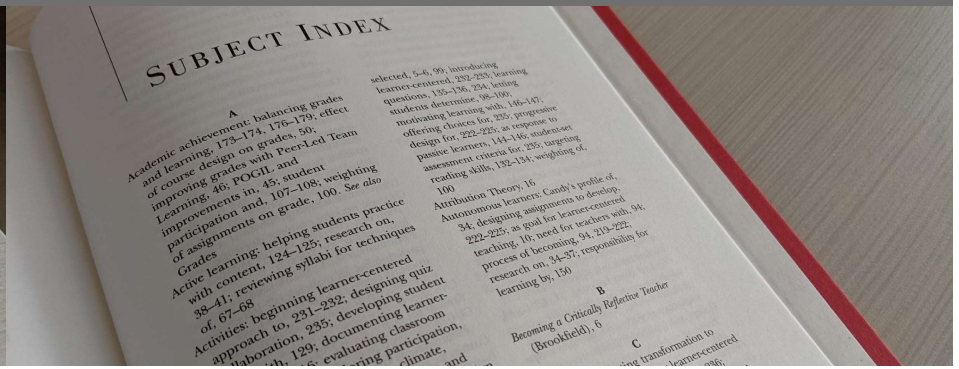
Sparse

- Maps some search keys.
- Use index and linear search.
- Table sorted on search key.
- Smaller

Dense

- Maps all search keys.
- Use index and done.





Covering index

Index alone is sufficient to answer query.

- Dense
- Search key contains all used attributes

prod_manufacturer	refs
NULL	
GizmoWorks	
Hyper	
WidgetsRUs	

```
SELECT prod_manufacturer
FROM Product
WHERE prod_manufacturer IS NOT NULL
```

Ordered index

Uses an ordering relation on the data.

Most “obvious” indices do.

Various index data structures

We'll focus on three:

- B+ trees
- Hash buckets
- Bitmap

Who chooses index?

DBMS automatically chooses!

- Based upon statistics & heuristics.

DB programmer able to choose.

- Able to see DBMS choice.
- Able to specify different choice.